

# Design and Simulation of PID Control Strategy for Quadrotor Drones Integrating Kinematics and Dynamics Modeling

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**Abstract**—Quadrotors are currently applied in agriculture, logistics, photography, rescue, and other areas. But quadrotors are unstable nonlinear systems, translation and rotation are highly coupled, and it is very difficult to control the flight accurately and stably. Especially when there is a disturbance such as wind or a sudden change in load. This paper aims to design and simulate the Proportional-Integral-Derivative control strategy for quadrotor drones based on kinematic and dynamic modeling. The research is based on the current demand for stable and accurate quadrotor flight with various operating conditions and external disturbances. Then, the kinematic and dynamic equations of the quadrotor are derived to describe its translation and rotation characteristics. A cascaded PID control structure is formulated, including inner-loop attitude control and outer-loop position control. The control gains are initially estimated using the Ziegler - Nichols method and then adjusted in a Simulink simulation. The simulation results show that the proposed PID strategy can maintain a satisfactory stabilization, fast response, and accurate tracking of the trajectory while suppressing oscillation and steady-state error. The results also show that accurate modeling combined with classical PID control can improve the flight precision, stability, and robustness of quadrotor drones significantly. The research work provides a preliminary basis for advanced nonlinear and adaptive control exploration.

**Keywords**—Mechanical engineering, UAV, drone, PID Control, automation, robot, Artificial Intelligence

## I. INTRODUCTION

In recent years quadrotor drones have accomplished notable success in fields like agriculture, logistics and transportation, aerial photography, public safety, emergency rescue and so on. A strong flight performance under different operating conditions and disturbances can be accomplished by incorporating accurate kinematic and dynamic models into the control strategy design. Classical Proportional-Integral-Derivative (PID) control is still the most widely used due to its simplicity and ease of realization and satisfactory performance in a wide variety of situations [1].

On the other hand, classical PID controllers lack of robustness, are sensitive to noise and present problems related to feedback or potential for integral windup when they are used in systems with nonlinear dynamics if they are not properly tuned or modeled [2].

To deal with the problem addressed in this work the

proposed approach starts with the derivation of the quadrotors kinematic and dynamic equations. Subsequently a classical PID controller is grown and tuned in order to control the drone position and attitude. Finally, the whole system is developed and simulated in Simulink where different test scenarios are considered in order to evaluate the controller behavior. When implemented over a fully-modeled quadrotor system, the results obtained through the simulation analysis in this work are of great importance in order to evaluate the effectiveness of the PID strategy and will pave the way for future works considering different advanced control strategies.

The main objective of this work is to develop and simulate a PID control strategy for quadrotor drones by means of kinematic and dynamic modeling. The combination shown here represents a more accurate behavior of the drone and shows that the controller is responding better to the physical drone behavior, increasing its stability, improving its response and finally achieving a higher flight precision.

## II. METHODS

### A. Kinematics

The kinematic representation of the quadrotor unmanned aerial vehicle (UAV) describes the motion variables. However, it doesn't consider the forces and moments responsible for their generation [3]. The translational kinematics are formulated in the inertial reference frame  $O_e, X_e, Y_e, Z_e$ , where the position vector is defined as  $\vec{P}=[x,y,z]^T$ .

The temporal evolution of position is given by:  $\dot{x} = V_x$ ,  $\dot{y} = V_y$ ,  $\dot{z} = V_z$ .

The rotational kinematics are expressed in the body-fixed frame  $O_b, X_b, Y_b, Z_b$ , in which the Euler angles.  $\phi$ ,  $\theta$ ,  $\psi$  correspond to roll, pitch, and yaw, respectively. The body angular rates are denoted by  $p$ ,  $q$ ,  $r$ . The relationship between the Euler angle rates and the body angular rates is expressed as :

$$\dot{\phi} = p + (r \cos \phi + q \sin \phi) \tan \theta \quad (1)$$

$$\dot{\theta} = q \cos \phi - r \sin \phi \quad (2)$$

$$\dot{\psi} = \frac{1}{\cos \theta} (r \cos \phi + q \sin \phi) \quad (3)$$

In this case , a “+” configuration quadrotor is considered,

with the  $X_b$  axis aligned with the forward rotor, the  $Y_b$  axis oriented to the right, and the  $Z_b$  axis directed upward, complying with the right-hand rule [1]. The Coordinate system and force analysis diagram of a quadrotor drone is shown in Figure 1.

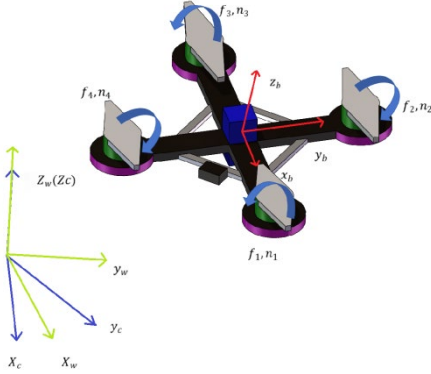


Fig. 1. Coordinate system and force analysis diagram of a quadrotor drone (Picture credit : Original )

### B. Dynamics

The UAV dynamic equations are established based on Newton's Second Law for translational motion and Euler's rotational equations for angular motion, assuming a rigid body and neglecting aerodynamic drag effects.

#### 1) Translational Dynamics

In the inertial frame, the translational motion is governed by:

$$m\dot{V}_x = F_t (\cos\theta\cos\psi\sin\theta + \sin\phi\sin\psi) \quad (4)$$

$$m\dot{V}_y = F_t (\cos\phi\sin\phi\sin\psi - \cos\psi\sin\theta) \quad (5)$$

$$m\dot{V}_z = -mg + F_t\cos\phi\cos\psi \quad (6)$$

where  $m$  is the UAV mass,  $g$  is the gravitational acceleration, and  $F_t = F_1 + F_2 + F_3 + F_4$  represents the total thrust generated by the four rotors [4].

#### 2) Rotational Dynamics

The rotational dynamics in the body-fixed frame are expressed as:

$$I_x\dot{p} = (I_y - I_z)qr + \tau_\phi \quad (7)$$

$$I_y\dot{q} = (I_z - I_x)pr + \tau_\theta \quad (8)$$

$$I_z\dot{r} = (I_x - I_y)pq + \tau_\psi \quad (9)$$

where  $I_x, I_y, I_z$  denote the principal moments of inertia, and  $\tau_\phi, \tau_\theta, \tau_\psi$  are the control moments generated by rotor thrust differences [2]:

$$\tau_\phi = l(F_4 - F_2) \quad (10)$$

$$\tau_\theta = l(F_3 - F_1) \quad (11)$$

$$\tau_\psi = d(-F_1 + F_2 - F_3 + F_4) \quad (12)$$

Here,  $l$  is the arm length from the rotor to the UAV center of mass, and  $d$  is the torque coefficient [5].

### C. Cascaded Control Architecture

The UAV controller is structured in a hierarchical manner:

Rate control (inner-most loop) regulates angular velocities  $(p, q, r)$ . It provides fast stabilization of rotational rates and is modeled as a black-box system with high bandwidth [10].

Attitude control (middle loop) regulates roll, pitch, and yaw angles  $(\phi, \theta, \psi)$ , using angular rate outputs as feedback to ensure stable orientation [10].

Position control (outer loop) regulates translational motion  $(x, y, z)$ . Position errors in the horizontal plane are mapped to reference roll and pitch angles, while altitude is controlled via collective thrust [1] [3].

This cascaded structure separates the fast inner-loop dynamics from the slower outer-loop dynamics, thus improving control stability and tuning efficiency [6] [10].

#### 1) PID Controller Formulation

The PID controller computes the control input  $u(t)$  based on the tracking error  $e(t) = r(t) - y(t)$ , where  $r(t)$  is the reference signal and  $y(t)$  is the measured output. The time-domain formulation is given by:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (13)$$

where  $K_p, K_i$  and  $K_d$  denote the proportional, integral, and derivative gains, respectively [7] [8].

The proportional term provides immediate correction in proportion to the error. The integral term eliminates steady-state error by accumulating past error over time. The derivative term predicts future error trends and improves damping characteristics [6][8].

Formulation framework is shown as below [3][10]: (based on cascaded UAV control loops)

Rate control (inner loop) controls angular rates  $(p, q, r)$ . For example, the roll rate controller is formulated as:

$$\tau_p(t) = K_p^r e_p(t) + K_i^r \int_0^t e_p(\tau) d\tau + K_d^r \frac{de_p(t)}{dt}, \quad (14)$$

Where

$$e_p(t) = p_{ref} - p. \quad (15)$$

Attitude control (middle loop) controls Euler angles  $(\phi, \theta, \psi)$ . The roll controller is expressed as:

$$\tau_\phi(t) = K_p^a e_\phi(t) + K_i^a \int_0^t e_\phi(\tau) d\tau + K_d^a \frac{de_\phi(t)}{dt}, \quad (16)$$

Where

$$e_\phi(t) = \phi_{ref} - \phi. \quad (17)$$

Position control (outer loop) controls translational motion  $(x, y, z)$ . For altitude regulation, the thrust command is:

$$F_z(t) = K_p^p e_z(t) + K_i^p \int_0^t e_z(\tau) d\tau + K_d^p \frac{de_z(t)}{dt}, \quad (18)$$

Where

$$e_z(t) = z_{ref} - z. \quad (19)$$

This hierarchical PID framework ensures that rate, attitude, and position loops are stabilized at different time scales. In practice, tuning is performed from the innermost rate loop outward to achieve overall system stability [3] [8] [10].

In practice, different forms of PID controllers can be applied in the areas mentioned.

- P control: fast response but large steady-state error.
- PI control: eliminates steady-state error but may induce oscillation.
- PID control: provides balanced transient and steady-state performance [9].

## 2) Control Strategy for UAV

For quadrotor control, the PID design is structured in a cascaded form:

Inner-loop attitude control regulates roll ( $\phi$ ), pitch ( $\theta$ ) and yaw ( $\psi$ ) angles using angular rate feedback (p,q,r). The control inputs are the required torques  $\tau_\phi$ ,  $\tau_\theta$  and  $\tau_\psi$ , which are mapped to rotor thrust differences [6] [7].

Outer-loop position control regulates translational motion ( $x,y,z$ ) by generating reference attitude commands

for the inner loop. For example, errors in the  $x$ - $y$  plane are converted into desired pitch and roll angles, while altitude  $z$  is controlled by the collective thrust.

This cascaded structure separates the fast attitude dynamics from the slower translational dynamics, improving system stability and simplifying parameter tuning.

## 3) Gain Tuning Strategy

The gains  $K_p$ ,  $K_i$  and  $K_d$  are selected using a combination of theoretical and empirical methods. The Ziegler–Nichols second method provides an initial estimation by increasing  $K_p$  until sustained oscillation occurs, after which  $K_i$  and  $K_d$  are introduced according to heuristic rules [8]. In practice, fine-tuning is performed iteratively in Simulink simulations to achieve desirable transient performance (rise time, settling time, overshoot) while ensuring adequate stability margins [6] [8].

## 4) Practical Considerations

Integral windup is a common issue when actuators saturate, leading to degraded response. Anti-windup schemes are adopted by limiting the integral term within a predefined range. In addition, the control allocation stage transforms the total thrust and torques ( $F_t, \tau_p, \tau_q, \tau_r$ ) into individual rotor thrusts ( $F_1, F_2, F_3, F_4$ ) via an allocation matrix, ensuring coordinated translational and rotational motion [8] [10].

## III. RESULTS

The detailed structure of the control system of one UAV is shown in Figure 2:

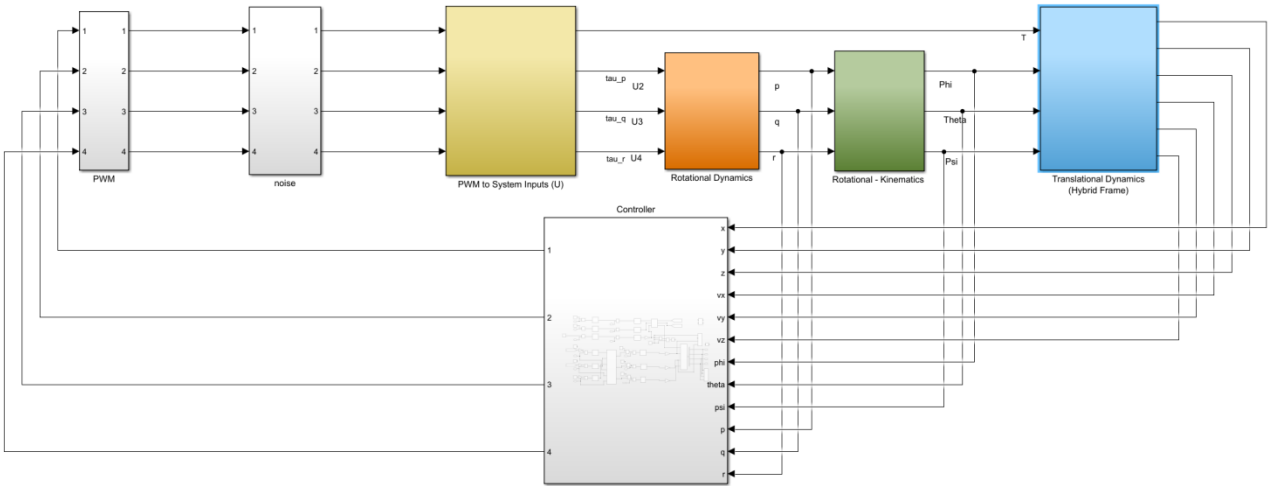


Fig. 2. Structure of the Control System of One UAV(Picture credit : Original )

Rotational Dynamics Simulation is shown in Figure 3:

Figure 3 indicate that attitude stabilization was tested for roll, pitch, and yaw, where the PID controller effectively

reduced oscillations and achieved smooth convergence, while also demonstrating robustness against small disturbances.

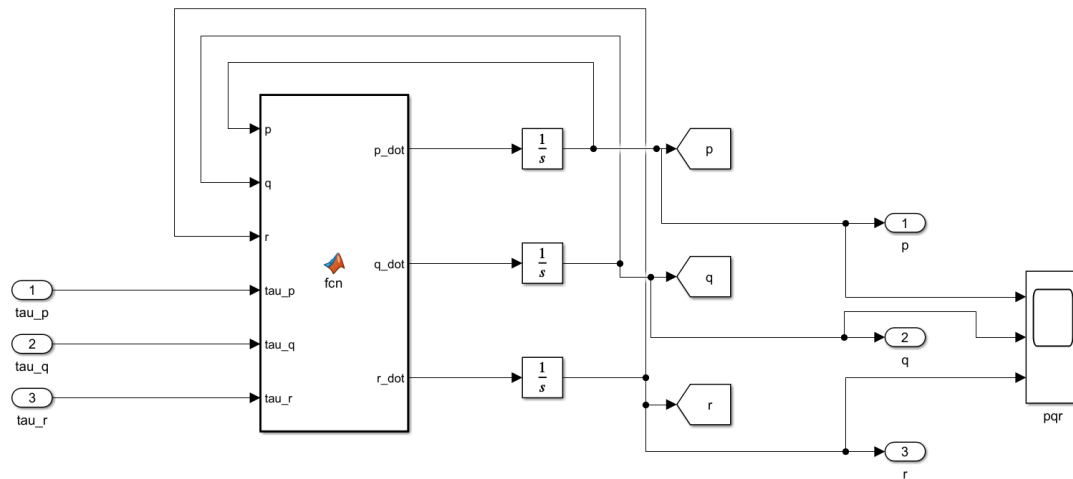


Fig. 3. Rotational Dynamics Simulation(Picture credit : Original )

Rotational Kinematics Simulation is shown in figure 4:

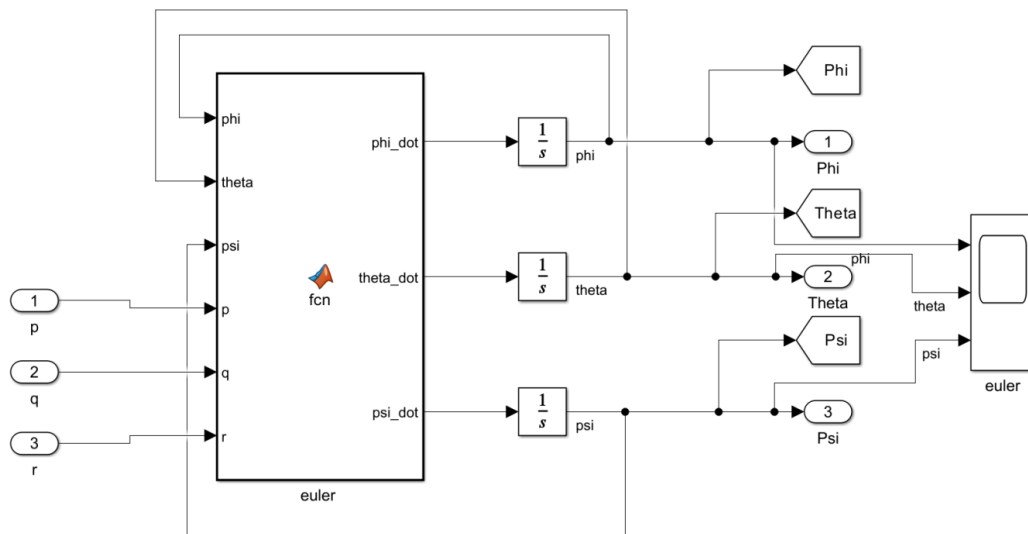


Fig. 4. Rotational Kinematics Simulation(Picture credit : Original )

Figure 4 shows the simulation of Euler angles ( $\phi$ ,  $\theta$ ,  $\psi$ ) based on the kinematic equations verified the intrinsic relationship between the body angular rates ( $p$ ,  $q$ ,  $r$ ) and the corresponding attitude variations, thereby ensuring

consistency between the theoretical formulation and the observed dynamic behavior.

Translational Dynamics Simulation is shown in figure 5:

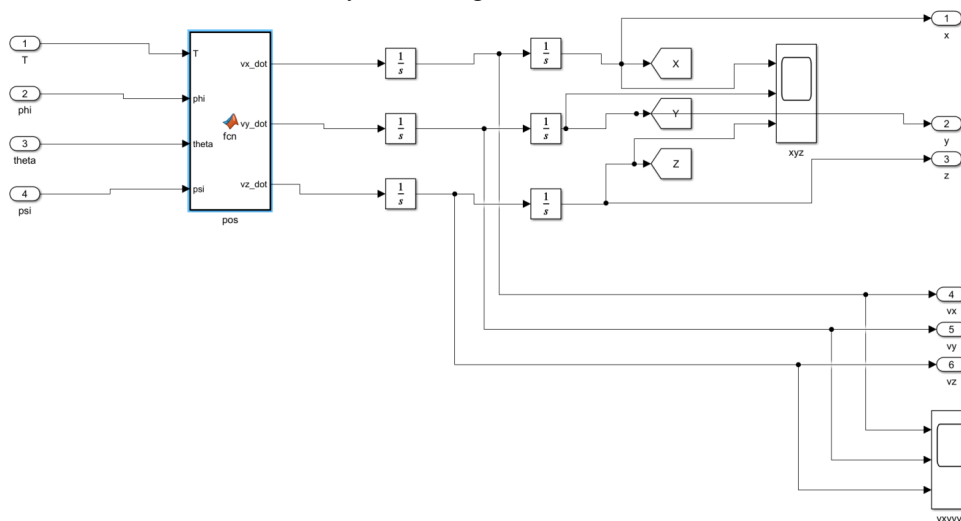


Fig. 5. Translational Dynamics Simulation (Picture credit : Original )

The Translational Dynamics Simulation Results is shown in Figure 6 and Figure 7:

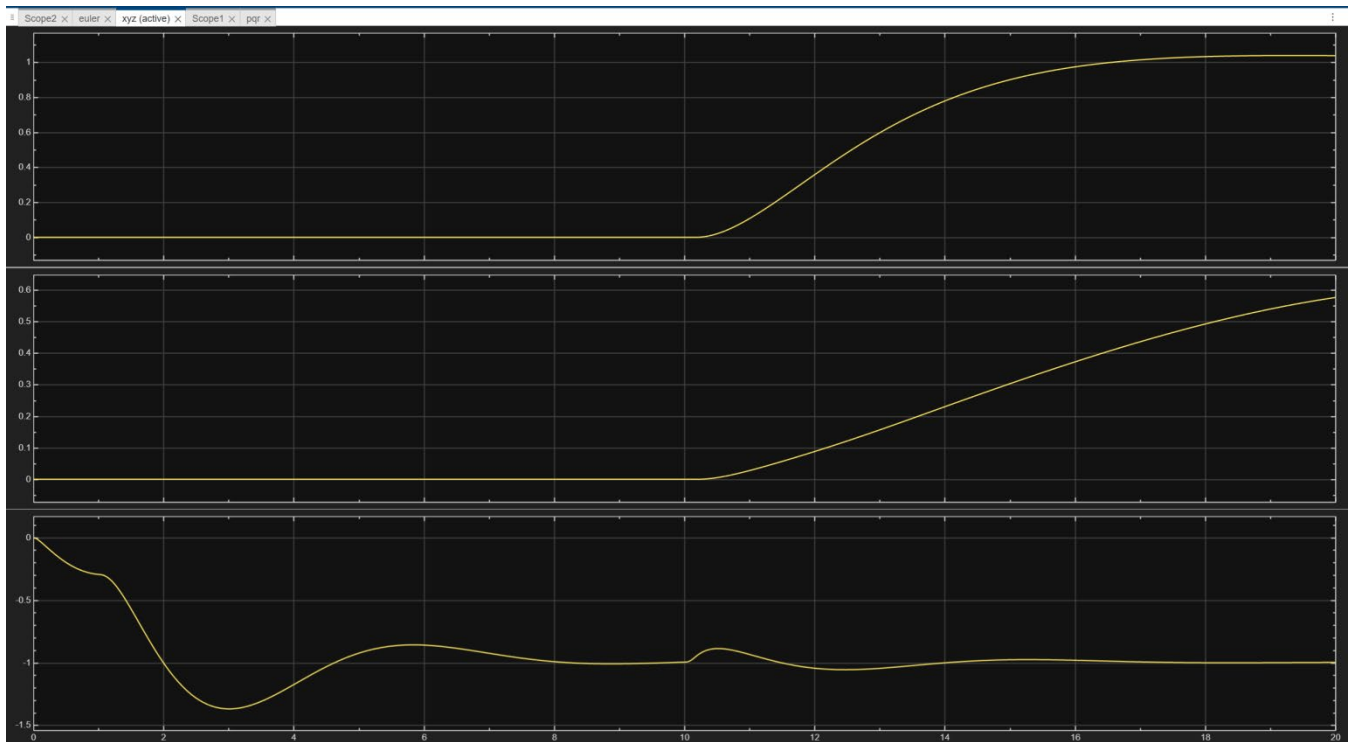


Fig. 6. Translational Dynamics Simulation Results (Picture credit : Original )

Figure 6 illustrates the time-domain responses of the quadrotor states under the proposed cascaded PID control strategy. The figure is divided into three subplots. The top subplot represents the evolution of a key system state, which exhibits a smooth increase after a short delay and eventually converges to the reference value. The middle subplot shows the response of another state variable, which grows gradually and demonstrates stable tracking behavior. The

bottom subplot depicts the error dynamics, where oscillations are observed in the initial transient phase but gradually decay, indicating that the controller successfully suppresses disturbances and ensures convergence. Together, these results confirm that the designed PID controller is capable of stabilizing the quadrotor dynamics and achieving satisfactory tracking performance.

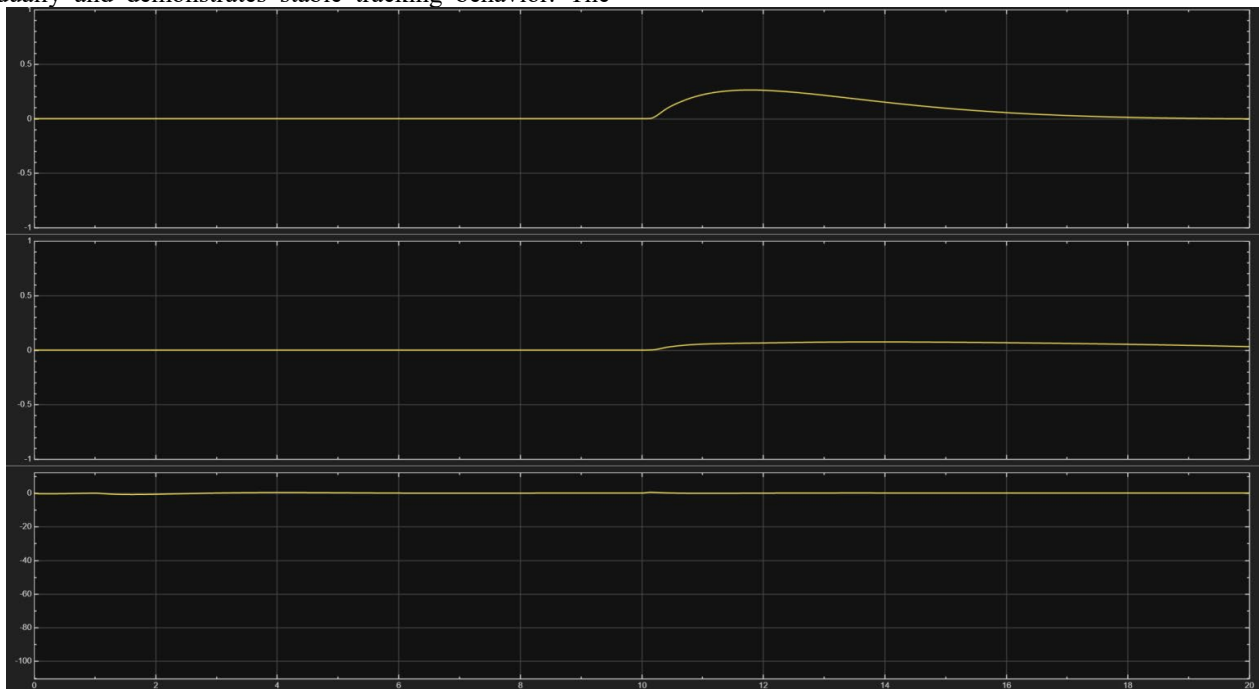


Fig. 7. Translational Dynamics Simulation Results (Picture credit : Original )

Figure 7 presents the attitude and control input responses to a step reference command. The top subplot displays the angular response, which experiences a limited overshoot

before returning smoothly towards stability. The middle subplot represents another angular or velocity component, which follows the reference input and stabilizes within an

acceptable settling time. The bottom subplot shows the control input signals, demonstrating that the commanded torques remain bounded and do not exhibit excessive oscillations. This confirms that the PID design, supported by

anti-windup compensation, maintains stability while preventing actuator saturation.

PID control Results are shown in Figure 8 and Figure 9:

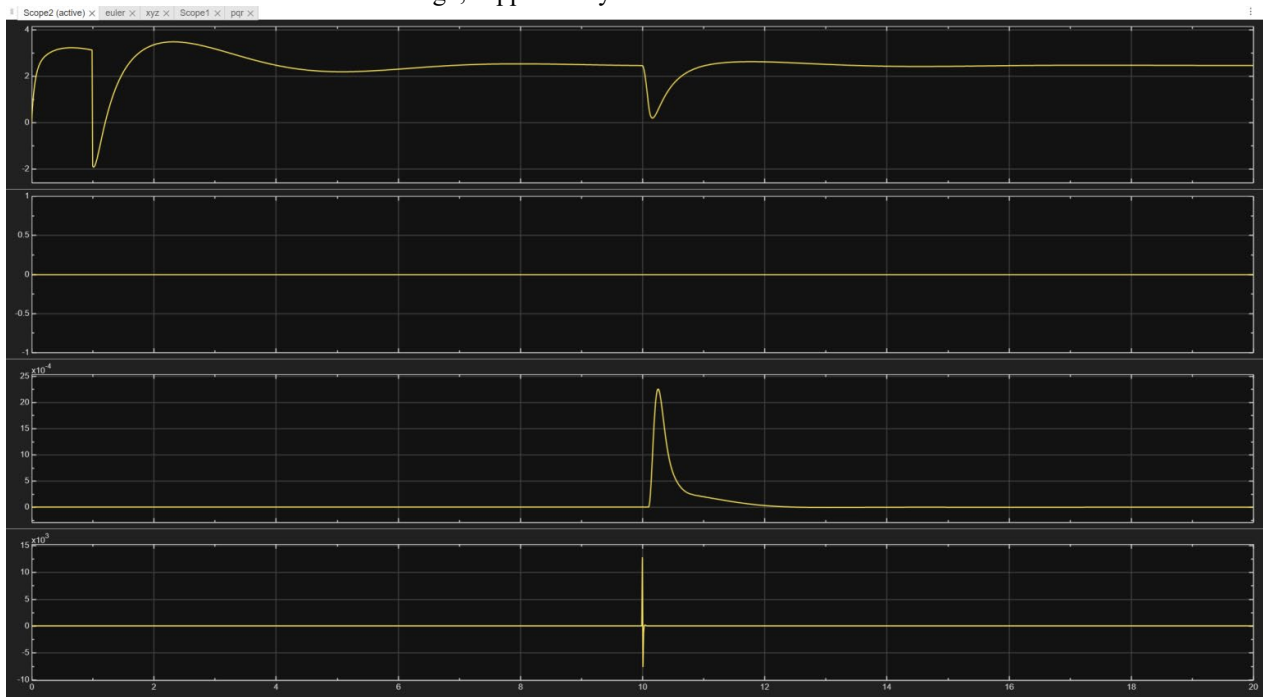


Fig. 8. PID control Results (Picture credit : Original )

Figure 8 shows the system responses under the cascaded PID controller during a reference tracking and disturbance rejection scenario. The figure is divided into four subplots. The top subplot illustrates the primary state variable response, which initially exhibits overshoot and oscillations but subsequently stabilizes at the desired value. This behavior indicates the effectiveness of the proportional and derivative actions in suppressing transient deviations. The second subplot represents a secondary state that remains

nearly constant, reflecting that the controller successfully isolates cross-coupling effects. The third subplot shows the error dynamics on a logarithmic scale, where a transient peak occurs at the moment of input change but rapidly decays, confirming integral action compensation. The fourth subplot demonstrates another error-related metric, which also converges quickly to zero, further validating the robustness of the PID design.

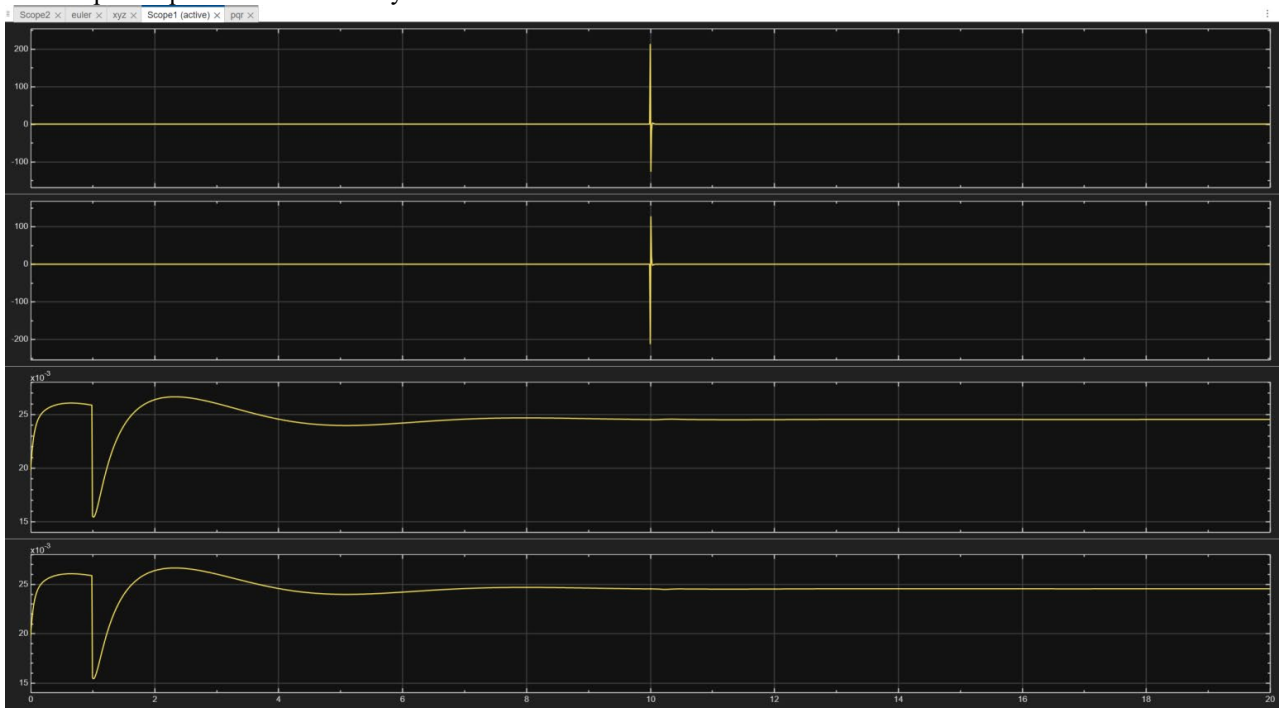


Fig. 9. PID control Results (Picture credit : Original )

Figure 9 presents the corresponding control inputs and additional dynamic responses of the quadrotor. The first and second subplots display the actuator control signals, which experience short-duration spikes at the moment of reference changes but remain bounded afterward, thereby avoiding actuator saturation. The third subplot shows the time-domain response of a state variable in a logarithmic scale, highlighting the suppression of high-frequency oscillations by derivative damping. Finally, the fourth subplot demonstrates another stability-related state, which converges smoothly, confirming that the cascaded PID controller achieves a stable and well-damped closed-loop system.

#### IV. DISCUSSION

Three points are summarized as follows. First, the complete kinematic and dynamic modeling framework of the quadrotor was constructed. Second, the cascaded PID control strategy was designed and tuned in a systematic way to guarantee the stable regulation of attitude and position loops. Third, the validity of the whole approach was further checked in Simulink simulations, and satisfactory tracking and stabilization performance under nominal conditions were obtained.

There are still some limitations of the work. The modeling and control design are carried out under ideal circumstances without aerodynamic drag, wind disturbance, and sensor noise. These assumptions limit the application of the above results to practical situations where the above disturbances exist.

How to proceed with the above limitations will be the biggest issue. In the future, robust and adaptive control schemes for uncertainties will be studied. In addition, intelligent control, such as fuzzy control and neural networks, will be used to improve the adaptability.

#### V. CONCLUSION

In this work, a PID-based control scheme for quadrotor

drones was designed and evaluated using kinematic and dynamic models. The cascaded structure decoupled the fast rotational dynamics from the slower translational dynamics, which allowed for proper tuning and stable running. Simulations in Simulink showed that the controller achieved accurate position and attitude regulation, and had good transient performance as well as acceptable robustness against typical disturbances.

Classical PID control is powerless in eliminating the impacts induced by strong nonlinearities and uncertainties. However, this work has attempted to use it as a simple baseline for quadrotor control. It is recommended that adaptive or robust/intelligent control be explored in future works.

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